Acute Heat Stress and Cranberry Yield

Executive Summary

Across the 2024 and 2025 acute heat experiments, heat stress reduced both total and healthy fruit weight. The analysis accounted for the paired plots and adjusted for differences between years and cultivar fields. This reduces shared background differences but does not remove all confounding. Averaged over the four intensity levels, both years, and both cultivars, the fitted model puts total weight about 12% below controls, and healthy weight about 20% below (95% interval roughly 14–25%). The healthy part of the harvest also fell, so the losses were not only in total weight but also in the composition of the crop.

The largest estimated losses were in 2025 and at the highest intensity. In 2025, the four-pulse treatment was about 37% below control for total weight and about 40% below for healthy weight. The 2025 average loss was larger than in 2024. It is about 15% versus 8% for total weight, but the size of the year difference remains uncertain. Across the four intensity levels, total weight fell by about 9% per step overall and about 15% per step in 2025.

The two healthy part measures moved in the same direction. The count of healthy part was about 11 percentage points lower than in controls; the weight part was about 7 points lower. These are changes in composition, which is a smaller part of the harvest was healthy, consistent with a higher rot part.

Heat treatment reduced healthy fruit weight more than total fruit weight, meaning that total harvest weight alone understates the loss of healthy fruit. The clearest losses occurred under the four-pulse treatment in 2025, when total weight was about 37% below controls and healthy weight was about 40% below. The intensity pattern was weaker in 2024, so the results do not support a fixed loss for every additional pulse. Neither cultivar showed a clearly different proportional response to heating. For growers, these findings support tracking healthy fruit weight and rot share alongside total yield when assessing heat damage.

Contributions

Author contributions are listed in the table below.

Author	Contribution	Share of the work
Soren Basnet	Ttechnical report，presentation	Report section 1, 2, 5, 6
Qingquan Liu	Technical Report，presentation	Report section 1，2，4 & appendix B
Wuyang Wang	Tecnical Report, Executive Summary， presentation	Report section 1, 2, 3 & appendix A

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